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Inventor(s)

KWON; Munseong et al.

ROTARY COMPRESSOR HAVING FLAT MUFFLER

Abstract

A rotary compressor includes: a casing; a motor disposed inside the casing; a compression part disposed below the motor; a lower flange disposed below the compression part; a muffler, having a portion that is flat, disposed on a lower surface of the lower flange; and a sealing plate disposed between the muffler and the lower flange.

Inventors: KWON; Munseong (Suwon-si, KR), KIM; Joonhyung (Suwon-si, KR), PARK; Jaewoo (Suwon-si, KR), LEE; Jeongbae (Suwon-si, KR), CHOI; Jongwon (Suwon-si, KR)

Applicant: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

Family ID: 91085079

Assignee: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application is a continuation application, under 35 U.S.C. § 111(a), of international application No. PCT/KR2023/016758, filed Oct. 26, 2023, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2022-0155777, filed Nov. 18, 2022, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The disclosure relates to a rotary compressor, and more particularly, to a rotary compressor having a flat muffler.

BACKGROUND ART

[0003] A compressor is a mechanical device that compresses gas to increase pressure, and may be classified into reciprocating compressors and rotating compressors based on their operating principles.

[0004] The reciprocating compressor may include a recipro compressor that converts the rotational motion of a motor into the linear reciprocating motion of a piston using a crank shaft and a connecting rod to suck and compress gas.

[0005] The rotating compressor may include a rotary compressor and a scroll compressor.

[0006] The rotary compressor may be configured so that refrigerant is sucked and compressed by a roller rotating inside a cylinder of a compression part by the rotational motion of a motor.

[0007] The scroll compressor may be configured so that refrigerant is sucked and compressed by an orbiting scroll rotating in a certain direction relative to a fixed scroll by the rotational motion of a motor.

[0008] In the rotary compressor, the compressed refrigerant may be discharged inside a casing and discharged to the outside through a refrigerant discharge pipe together with oil.

DISCLOSURE OF INVENTION

Technical Solution

[0009] According to an aspect of the disclosure, a rotary compressor may include: a casing; a motor disposed inside the casing; a compression part disposed below the motor; a lower flange disposed below the compression part; a muffler, having a portion that is flat, to be disposed on a lower surface of the lower flange; and a sealing plate disposed between the flat muffler 72 and the lower flange.

[0010] The portion that is flat is an upper end of the flat muffler may not protrude above the lower surface of the lower flange.

[0011] The muffler may be formed in a dome shape. The muffler may include a plurality of concave portions formed at regular intervals along a circumferential direction of the muffler on a side surface of the muffler. The sealing plate may be formed in a ring shape. An inner circumferential surface of the sealing plate may be formed concavely and convexly corresponding to the muffler.

[0012] The lower flange may include a plurality of lower refrigerant holes and a plurality of first bolt holes formed along an edge of the lower flange. The muffler may include a plurality of second bolt holes formed corresponding to the plurality of first bolt holes of the lower flange at positions corresponding to the plurality of concave portions along an edge of the muffler.

[0013] The sealing plate may include a plurality of bolt seats formed to protrude toward a center of the sealing plate from an inner circumferential surface of the sealing plate and having a plurality of third bolt holes corresponding to the plurality of second bolt holes of the muffler.

[0014] The sealing plate may be formed so as not to cover the plurality of lower refrigerant holes of the lower flange.

[0015] A minimum distance between the inner circumferential surface of the sealing plate **100** and a plurality of lower refrigerant holes of the lower flange may be zero (0).

[0016] A thickness of the sealing plate may have a following relationship with a thickness of the muffler: $0.5 \leq \text{thickness } T_s \text{ of the sealing plate } \mathbf{100} / \text{thickness } T_m \text{ of the muffler } \mathbf{72} \leq 2.0$.

[0017] Oil may be received in a lower portion of the casing, and the muffler may be immersed in the oil.

[0018] The sealing plate may be formed of one of heat-resistant resin, steel, copper, and a material in which at least two of the heat-resistant resin, the steel, and the copper are laminated.

[0019] The lower flange may include: a flange part formed in a disk shape; a boss extending vertically from the flange part and including a through hole; and a bearing disposed in the through hole of the boss. The muffler may include: a fixing plate formed in a ring shape and fixed to the flange part of the lower flange; a caulking portion fixed to one end of the boss; and a muffler portion provided between the fixing plate and the caulking portion.

[0020] The muffler portion may include a plurality of concave portions formed at regular intervals in a circumferential direction of the muffler.

[0021] The fixing plate may include a plurality of protrusions corresponding to the plurality of concave portions of the muffler portion.

[0022] The flange part of the lower flange may include a plurality of lower refrigerant holes and a plurality of first bolt holes. The plurality of protrusions of the fixing plate may include a plurality of second bolt holes corresponding to the first bolt holes.

[0023] An inner circumferential surface of the sealing plate may include a plurality of bolt seats formed to correspond to the plurality of protrusions of the fixing plate and a plurality of refrigerant grooves formed so as not to cover the plurality of lower refrigerant holes of the flange part.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0024] These and/or other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0025] FIG. **1** is a perspective view illustrating a rotary compressor according to an embodiment of the disclosure.

[0026] FIG. **2** is a cross-sectional view illustrating the rotary compressor of FIG. **1** taken along line A-A according to an embodiment of the disclosure.

[0027] FIG. **3** is a cross-sectional view illustrating the rotary compressor of FIG. **1** taken along line B-B according to an embodiment of the disclosure.

[0028] FIG. **4** is a perspective view illustrating a compression part of a rotary compressor according to an embodiment of the disclosure.

[0029] FIG. **5** is a bottom view illustrating the compression part of the rotary compressor of FIG. **4** according to an embodiment of the disclosure.

[0030] FIG. **6** is a cross-sectional view illustrating the compression part of the rotary compressor of FIG. **4** according to an embodiment of the disclosure.

[0031] FIG. **7** is an exploded perspective view illustrating the compression part of the rotary compressor of FIG. **4** according to an embodiment of the disclosure.

[0032] FIG. **8** is a perspective view illustrating a flat muffler used in a rotary compressor according to an embodiment of the disclosure.

[0033] FIG. **9** is a perspective view illustrating the flat muffler of FIG. **8** as seen from above

according to an embodiment of the disclosure.

[0034] FIG. **10** is a perspective view illustrating a sealing plate used in a rotary compressor according to an embodiment of the disclosure.

[0035] FIG. **11** is a view illustrating a state in which a sealing plate and a flat muffler are disposed in a lower flange according to an embodiment of the disclosure.

[0036] FIG. **12** is a bottom view of FIG. **11** according to an embodiment of the disclosure.

[0037] FIG. **13** is a bottom view illustrating a state in which the flat muffler is removed from FIG. **11** according to an embodiment of the disclosure.

[0038] FIG. **14** is a view illustrating a state in which a portion of a flat muffler is opened by compressed refrigerant according to an embodiment of the disclosure.

[0039] FIG. **15** is a graph illustrating a change in deformation of a bearing and opening of a flat muffler according to a thickness of a sealing plate according to an embodiment of the disclosure.

[0040] FIG. **16** is a graph illustrating a change rate in energy efficiency of a rotary compressor according to an embodiment of the disclosure relative to the energy efficiency of a rotary compressor according to the prior art using a skirt muffler.

BEST MODE FOR CARRYING OUT THE INVENTION

[0041] Since the embodiments of the disclosure can apply various transformations and have various embodiments, specific embodiments will be illustrated in the drawings and described in detail in the detailed description. However, this is not intended to limit the scope to the specific embodiments, and should be understood to include various modifications, equivalents, and/or alternatives of the embodiment of the disclosure. In connection with the description of the drawings, like reference numerals may be used for like elements.

[0042] In describing the disclosure, when it is determined that a detailed description of a related known function or configuration may unnecessarily obscure the gist of the disclosure, a detailed description thereof will be omitted.

[0043] In addition, the following embodiments may be modified in many different forms, and the scope of the technical idea of the disclosure is not limited to the following embodiments. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the spirit of the disclosure to those skilled in the art.

[0044] Terms used in this disclosure are only used to describe specific embodiments, and are not intended to limit the scope of rights. Singular expressions include plural expressions unless the context clearly dictates otherwise.

[0045] In this disclosure, expressions such as “has,” “can have”, “includes,” or “can include” indicate the existence of a corresponding feature (e.g., numerical value, function, operation, or component such as a part) and do not preclude the existence of additional features.

[0046] In this disclosure, expressions such as “A or B,” “at least one of A or/and B,” or “one or more of A or/and B” may include all possible combinations of the items listed together. For example, “A or B,” “at least one of A or/and B,” or “one or more of A or/and B” may refer to all cases (1) including at least one A, (2) including at least one B, or (3) including both at least one A and at least one B.

[0047] Expressions such as “first,” “second,” “primary,” or “secondary,” as used in this disclosure may modify various components regardless of order and/or importance, are used only to distinguish one component from other components, and do not limit the corresponding components.

[0048] Further, terms such as ‘leading end’, ‘rear end’, ‘upper side’, ‘lower side’, ‘top end’, ‘bottom end’, etc. used in the disclosure are defined with reference to the drawings. However, the shape and position of each component are not limited by these terms.

[0049] Hereinafter, embodiments of a rotary compressor according to the disclosure will be described in detail with reference to the attached drawings.

[0050] FIG. **1** is a perspective view illustrating a rotary compressor **1** according to an embodiment

of the disclosure.

[0051] Referring to FIG. 1, a rotary compressor **1** according to an embodiment of the disclosure may include a casing **10**.

[0052] The casing **10** may form an outer appearance of the rotary compressor **1**. The casing **10** may be configured as a sealed container. The casing **10** may include a refrigerant inlet **13** through which refrigerant is introduced and a refrigerant discharge pipe **14** through which refrigerant is discharged.

[0053] The rotary compressor **1** may form a refrigeration cycle together with a condenser, an expansion valve, and an evaporator. In this case, the refrigerant inlet **13** may be connected to the evaporator, and the refrigerant discharge pipe **14** may be connected to the condenser.

[0054] An accumulator **3** may be disposed on one side of the rotary compressor **1**. In this case, the refrigerant inlet **13** may be connected to the accumulator **3**. The inlet pipe of the accumulator **3** may be connected to the evaporator. Therefore, the refrigerant discharged from the evaporator may be introduced into the rotary compressor **1** through the accumulator **3**.

[0055] The casing **10** may include an upper casing **11** and a lower casing **12**. The upper casing **11** may be connected to the upper end of the lower casing **12** to form the casing **10**.

[0056] The joint of the upper casing **11** and the lower casing **12** may be sealed.

[0057] The refrigerant discharge pipe **14** may be provided in the upper casing **11**. The refrigerant discharge pipe **14** may be provided at the upper end of the upper casing **11**.

[0058] The refrigerant inlet **13** may be provided at the lower casing **12**. The refrigerant inlet **13** may be connected to a compression part **40** disposed inside the lower casing **12**. A low-temperature/low-pressure refrigerant may be introduced into the refrigerant inlet **13**. Therefore, the refrigerant may be introduced into the compression part **40** through the refrigerant inlet **13**.

[0059] The accumulator **3** may be disposed in the lower casing **12**. In this case, the refrigerant inlet **13** may be connected to the discharge pipe of the accumulator **3**.

[0060] A base **15** supporting the casing **10** may be provided at the bottom of the lower casing **12**. The rotary compressor **1** may be disposed vertically to the support surface by the base **15**.

[0061] FIG. 2 is a cross-sectional view illustrating the rotary compressor **1** of FIG. 1 taken along line A-A. FIG. 3 is a cross-sectional view illustrating the rotary compressor **1** of FIG. 1 taken along line B-B.

[0062] Referring to FIGS. 2 and 3, the rotary compressor **1** according to an embodiment of the disclosure may include the casing **10**, a motor **20**, and the compression part **40**.

[0063] The casing **10** may form the outer appearance of the rotary compressor **1**, and be a cylindrical sealed container. The casing **10** may include the lower casing **12** provided with the refrigerant inlet **13** and the upper casing **11** provided with the refrigerant discharge pipe **14**.

[0064] The casing **10** may be configured by connecting the upper casing **11** and the lower casing **12**, and the interior of the casing **10** except for the refrigerant inlet **13** and the refrigerant discharge pipe **14** may be sealed. In other words, the refrigerant may be introduced into the interior of the casing **10** or discharged to the outside from the casing **10** only through the refrigerant inlet **13** and the refrigerant discharge pipe **14**.

[0065] The internal space of the casing **10** may accommodate high-pressure refrigerant discharged from the compression part **40**, and the high-pressure refrigerant may be discharged to the outside through the refrigerant discharge pipe **14**.

[0066] An oil reservoir **16** for receiving oil may be provided at the lower portion of the casing **10**.

[0067] The accumulator **3** may be disposed on the outer surface of the casing **10**. At this time, the refrigerant inlet **13** may be connected to the discharge pipe of the accumulator **3**.

[0068] The motor **20** may be arranged in the upper side inside the casing **10**. The motor **20** may include a stator **21** and a rotor **22**.

[0069] The stator **21** of the motor **20** may be fixed to the inner circumferential surface of the casing **10**. A plurality of oil return passages may be provided between the outer circumferential surface of

the stator **21** and the inner circumferential surface of the casing **10**. The plurality of oil return passages may be formed at regular intervals along the outer circumferential surface of the stator **21**. [0070] Oil from the upper side of the motor **20** may move to the lower side of the motor **20** through the plurality of oil return passages provided between the stator **21** and the casing **10**. Oil that has moved to the lower side of the motor **20** may be collected in the oil reservoir **16** provided at the lower portion of the casing **10**.

[0071] The rotor **22** may be rotatably disposed at the center of the stator **21**. The rotor **22** may be disposed so as to maintain a certain gap with the inner surface of the stator **21**.

[0072] A shaft hole **29** may be provided at the center of the rotor **22** that penetrates the rotor **22** in the longitudinal direction. A plurality of refrigerant holes **27** may be provided around the shaft hole **29** of the rotor **22**. The plurality of refrigerant holes **27** may be formed to penetrate the rotor **22** in the longitudinal direction, that is, in the up-down direction.

[0073] The refrigerant discharged from the compression part **40** below the motor **20** may move to the upper side of the motor **20** through the gap between the rotor **22** and the stator **21** and the plurality of refrigerant holes **27**.

[0074] A drive shaft **30** may be inserted into and fixed in the shaft hole **29** penetrating the center of the rotor **22**. Therefore, when power is applied to the motor **20**, the rotor **22** may rotate by the electromagnetic force acting between the stator **21** and the rotor **22**, and the drive shaft **30** may rotate integrally with the rotor **22**.

[0075] When the drive shaft **30** rotates by the motor **20**, the compression part **40** may operate to compress the refrigerant.

[0076] The drive shaft **30** may be formed to extend downward from the motor **20**. The lower portion of the drive shaft **30** extending downward from the motor **20** may be connected to the compression part **40**. The lower portion of the drive shaft **30** may be configured as a crank shaft to operate the compression part **40**.

[0077] The crank shaft of the drive shaft **30** may include two eccentric portions, that is, an upper eccentric portion **31** and a lower eccentric portion **32**. Therefore, when the drive shaft **30** rotates, the upper eccentric portion **31** and the lower eccentric portion **32** of the crank shaft may rotate integrally with the drive shaft **30**.

[0078] The upper eccentric portion **31** may be formed in a cylindrical shape having a diameter larger than the diameter of the drive shaft **30**. The center line of the upper eccentric portion **31** may be eccentric with the center line of the drive shaft **30**. An upper roller **33** may be disposed on the outer circumferential surface of the upper eccentric portion **31**.

[0079] The lower eccentric portion **32** may be disposed below the upper eccentric portion **31**, and may be formed in the same manner as the upper eccentric portion **31**. In other words, the lower eccentric portion **32** may be formed in a cylindrical shape having a diameter larger than the diameter of the drive shaft **30**. The center line of the lower eccentric portion **32** may be eccentric with the center line of the drive shaft **30**. A lower roller **34** may be disposed on the outer circumferential surface of the lower eccentric portion **32**.

[0080] The lower eccentric portion **32** may be formed to be eccentric in a different direction from the upper eccentric portion **31** with respect to the center line of the drive shaft **30**. For example, the lower eccentric portion **32** may be eccentric 180 degrees opposite to the upper eccentric portion **31** with respect to the center line of the drive shaft **30**.

[0081] The drive shaft **30** may be rotatably supported by an upper flange **91** and a lower flange **92**. The upper flange **91** may be disposed to support the drive shaft **30** between the rotor **22** and the upper eccentric portion **31**. The lower flange **92** may be disposed to support the lower end portion of the drive shaft **30** below the lower eccentric portion **32**.

[0082] The compression part **40** may be disposed below the motor **20**. The compression part **40** may be configured to compress the refrigerant and discharge the compressed refrigerant to the upper side of the compression part **40** according to the rotation of the drive shaft **30**.

[0083] Hereinafter, the compression part **40** of the rotary compressor **1** according to an embodiment of the disclosure will be described in detail with reference to FIGS. **4** to **7**.

[0084] FIG. **4** is a perspective view illustrating a compression part **40** of a rotary compressor **1** according to an embodiment of the disclosure. FIG. **5** is a bottom view of the compression part **40** of the rotary compressor **1** of FIG. **4**. FIG. **6** is a cross-sectional view illustrating the compression part **40** of the rotary compressor **1** of FIG. **4**. FIG. **7** is an exploded perspective view illustrating the compression part **40** of the rotary compressor **1** of FIG. **4**.

[0085] The compression part **40** may be disposed at the lower portion of the casing **10**, and may be configured to suck, compress, and discharge refrigerant by operating by the drive shaft **30** that is rotated by the motor **20**.

[0086] Referring to FIGS. **4** to **7**, the compression part **40** may be disposed between the upper flange **91** and the lower flange **92** that rotatably support the drive shaft **30**.

[0087] The compression part **40** may include an upper compression part **41**, a lower compression part **42**, and an intermediate plate **70** provided between the upper compression part **41** and the lower compression part **42**.

[0088] The upper compression part **41** may be configured to suck and compress the refrigerant according to the rotation of the drive shaft **30**. The lower compression part **42** may be provided below the upper compression part **41** and may be configured to suck and compress the refrigerant according to the rotation of the drive shaft **30**.

[0089] The upper compression part **41** may be disposed on the upper surface of the intermediate plate **70** and may include an upper cylinder **50** having a flat plate shape. The upper cylinder **50** may include a compression chamber. The compression chamber may be formed as a hollow **51** having a circular cross-section.

[0090] The upper roller **33** disposed in the upper eccentric portion **31** of the drive shaft **30** may be accommodated and rotated in the hollow **51** of the upper cylinder **50**.

[0091] The upper compression part **41** may include a refrigerant inlet passage **52** connected to the refrigerant inlet **13** provided in the casing **10**. The refrigerant inlet passage **52** may be formed in the upper cylinder **50**.

[0092] The refrigerant inlet passage **52** may be formed as a through hole that connects the hollow **51** and the outer circumferential surface of the upper cylinder **50**. Therefore, the refrigerant may be introduced into the hollow **51** of the upper cylinder **50** through the refrigerant inlet **13** and the refrigerant inlet passage **52**.

[0093] The upper compression part **41** may include an upper discharge port **501** through which the compressed refrigerant is discharged. The upper discharge port **501** may be provided on the upper surface of the upper cylinder **50**.

[0094] When the upper roller **33** rotates by the drive shaft **30**, the refrigerant may be introduced into the hollow **51** of the upper cylinder **50** through the refrigerant inlet passage **52**, compressed by the upper roller **33**, and discharged through the upper discharge port **501**.

[0095] The lower compression part **42** may be disposed on the lower surface of the intermediate plate **70** and may have a lower cylinder **60** having a flat plate shape. The lower cylinder **60** may include a compression chamber. The compression chamber may be formed as a hollow **61** having a circular cross-section.

[0096] The lower roller **34** disposed in the lower eccentric portion **32** of the drive shaft **30** may be accommodated and rotated in the hollow **61** of the lower cylinder **60**.

[0097] The lower compression part **42** may include a refrigerant inlet passage **62** connected to the refrigerant inlet **13** provided in the casing **10**. The refrigerant inlet passage **62** may be formed in the lower cylinder **60**.

[0098] The refrigerant inlet passage **62** may be formed as a through hole that connects the hollow **61** and the outer circumferential surface of the lower cylinder **60**. Therefore, the refrigerant may be introduced into the hollow **61** of the lower cylinder **60** through the refrigerant inlet **13** and the

refrigerant inlet passage **62**.

[0099] The lower compression part **42** may include a lower discharge port through which the compressed refrigerant is discharged. The lower discharge port may be provided on the lower surface of the lower cylinder **60**. Therefore, the refrigerant compressed by the lower compression part **42** may be discharged below the lower compression part **42** through the lower discharge port. [0100] When the lower roller **34** rotates by the drive shaft **30**, the refrigerant may be introduced into the hollow **61** of the lower cylinder **60** through the refrigerant inlet passage **62**, compressed by the lower roller **34**, and discharged through the lower discharge port.

[0101] The low-pressure refrigerant may be supplied to the upper compression part **41** and the lower compression part **42** through the accumulator **3**.

[0102] The intermediate plate **70** may be disposed between the upper cylinder **50** and the lower cylinder **60**. The intermediate plate **70** may be formed in a flat plate shape. Accordingly, the lower cylinder **60**, the intermediate plate **70**, and the upper cylinder **50** may be laminated to form the compression part **40**. The lower cylinder **60**, the intermediate plate **70**, and the upper cylinder **50** may be integrated connected by a plurality of bolts **93**.

[0103] The upper flange **91** may be disposed on the upper surface of the upper cylinder **50**. The upper flange **91** may be fixed to the inner circumferential surface of the casing **10**. Therefore, when the upper cylinder **50** is fixed to the upper flange **91**, the upper cylinder **50** may be fixed to the casing **10**.

[0104] The upper flange **91** may be configured to rotatably support the drive shaft **30** and block the upper end of the hollow **51** of the upper cylinder **50**.

[0105] The upper flange **91** may be provided with an upper through-hole communicating with the upper discharge port **501** of the upper cylinder **50**. Accordingly, the refrigerant discharged through the upper discharge port **501** of the upper cylinder **50** may be discharged to the upper side of the upper flange **91** through the upper through-hole of the upper flange **91**.

[0106] The upper flange **91** may include an upper discharge valve **911** configured to open and close the upper through-hole. Therefore, the upper through-hole of the upper flange **91** may be opened and closed by the upper discharge valve **911**. When the refrigerant introduced into the upper cylinder **50** is compressed, the upper discharge valve **911** may be opened so that the refrigerant may be discharged to the upper side of the upper flange **91**.

[0107] An upper muffler **71** may be disposed on the upper side of the upper flange **91**. The upper muffler **71** may be configured to reduce noise generated by the refrigerant discharged through the upper through-hole of the upper flange **91**.

[0108] The upper flange **91** may include a plurality of bolt holes **912** provided along the circumferential direction of the upper flange **91**.

[0109] The upper flange **91** may include a plurality of upper refrigerant holes **913** provided along the circumferential direction of the upper flange **91**. The refrigerant discharged from the lower cylinder **60** may flow to the upper side of the upper flange **91** through the plurality of upper refrigerant holes **913**.

[0110] The upper muffler **71** may include a plurality of refrigerant openings **711** through which the refrigerant may pass. The refrigerant passing through the upper flange **91** may be discharged into the space between the motor **20** and the compression part **40** through the plurality of refrigerant openings **711** of the upper muffler **71**.

[0111] The upper muffler **71** may be provided with a plurality of bolt holes **712** corresponding to the plurality of bolt holes **912** of the upper flange **91** along the edge of the upper muffler **71**.

[0112] A plurality of openings **95** may be provided at the edge of the upper flange **91**. The plurality of openings **95** may be formed around the upper muffler **71** disposed on the upper flange **91**. The plurality of openings **95** may be formed to penetrate the upper flange **91** upwardly and downwardly. Oil may move to the oil reservoir **16** at the lower portion of the casing **10** through the plurality of openings **95**.

[0113] The upper flange **91** may include an upper flange part **915**, an upper boss **916**, and an upper bearing **917**.

[0114] The upper flange part **915** may be formed in a disk shape. The upper flange part **915** may be formed to cover the hollow **51** of the upper cylinder **50**. The upper flange part **915** may be formed to have a diameter corresponding to the inner circumferential surface of the casing **10**. Therefore, the upper flange **91** may be fixed to the inner circumferential surface of the casing **10**.

[0115] The upper flange part **915** may include the upper through-hole communicating with the upper discharge port **501** of the upper cylinder **50**. Therefore, the refrigerant discharged through the upper discharge port **501** of the upper cylinder **50** may be discharged to the upper side of the upper flange part **915** through the upper through-hole of the upper flange part **915**.

[0116] The upper discharge valve **911** may be provided in the upper through-hole of the upper flange part **915**. Accordingly, the upper through-hole of the upper flange part **915** may be opened and closed by the upper discharge valve **911**. When the refrigerant introduced into the upper cylinder **50** is compressed above a certain pressure, the upper discharge valve **911** may be opened so that the refrigerant may be discharged to the upper side of the upper flange part **915**.

[0117] The upper boss **916** may be vertically extended from the center of the upper flange part **915**. The upper boss **916** may be vertically extended upward from the upper flange part **915**. A through hole may be formed in the center of the upper boss **916**.

[0118] The upper bearing **917** may be disposed in the through hole of the upper boss **916** and may rotatably support the drive shaft **30**. Any type of bearing may be used as the upper bearing **917** as long as it can rotatably support the drive shaft **30**. In the case of this disclosure, a sliding bearing may be used as the upper bearing **917**.

[0119] The plurality of bolt holes **912** and the plurality of refrigerant holes **913** may be provided in the upper flange part **915** around the upper boss **916**. The plurality of openings **95** may be provided on the edge of the upper flange part **915** outside the plurality of bolt holes **912**.

[0120] The upper cylinder **50** may be provided with a plurality of tap holes **502** corresponding to the plurality of bolt holes **912** of the upper flange part **915**. In addition, the upper cylinder **50** may include a plurality of refrigerant holes **503** corresponding to the plurality of upper refrigerant holes **913** of the upper flange **91**.

[0121] When the plurality of bolts **94** are fastened to the plurality of tap holes **502** of the upper cylinder **50**, the upper muffler **71** and the upper flange **91** may be fixed to the upper cylinder **50** by the plurality of bolts **94**.

[0122] The noise of the refrigerant discharged through the upper through hole of the upper flange **91** may be reduced as the refrigerant passes through the interior of the upper muffler **71**, and the refrigerant may be discharged to the upper side of the upper muffler **71**, that is, to the space between the motor **20** and the compression part **40** through the plurality of refrigerant openings **711** of the upper muffler **71**.

[0123] The lower flange **92** may be disposed on the lower surface of the lower cylinder **60**. The lower flange **92** may be configured to rotatably support the lower end portion of the drive shaft **30** and block the lower end of the hollow **61** of the lower cylinder **60**.

[0124] The lower flange **92** may be provided with a lower through-hole **924** communicating with the lower discharge port of the lower cylinder **60**. Accordingly, the refrigerant discharged through the lower discharge port of the lower cylinder **60** may be discharged to the lower side of the lower flange **92** through the lower through-hole **924** of the lower flange **92**.

[0125] The lower flange **92** may include a lower discharge valve **921** configured to open and close the lower through hole **924**. Therefore, the lower through-hole **924** of the lower flange **92** may be opened and closed by the lower discharge valve **921**. When the refrigerant introduced into the lower cylinder **60** is compressed to a certain pressure or higher, the lower discharge valve **921** may be opened so that the refrigerant may be discharged to the lower side of the lower flange **92**.

[0126] A flat muffler **72** may be disposed on the lower side of the lower flange **92**. The flat muffler

72 may be configured to reduce noise generated by the refrigerant discharged through the lower through-hole **924** of the lower flange **92**.

[0127] In addition, the lower flange **92** may be configured so that the refrigerant discharged through the lower through-hole **924** is not discharged below the flat muffler **72**.

[0128] The lower flange **92** may include a plurality of first bolt holes **922** provided along the circumferential direction of the lower flange **92**.

[0129] The lower flange **92** may include a plurality of lower refrigerant holes **923** provided along the circumferential direction of the lower flange **92**. The refrigerant discharged from the lower through-hole **924** of the lower flange **92** may flow to the upper side of the lower flange **92** through the plurality of lower refrigerant holes **923**.

[0130] In detail, the plurality of lower refrigerant holes **923** of the lower flange **92** may be formed to coincide with the plurality of refrigerant holes **603** of the lower cylinder **60**, the plurality of refrigerant holes **703** of the intermediate plate **70**, the plurality of refrigerant holes **503** of the upper cylinder **50**, and the plurality of upper refrigerant holes **913** of the upper flange **91**.

[0131] Accordingly, the refrigerant discharged from the lower through hole **924** of the lower flange **92** may move to the space formed by the upper flange **91** and the upper muffler **71** through the plurality of lower refrigerant holes **923** of the lower flange **92**, the plurality of refrigerant holes **603** of the lower cylinder **60**, the plurality of refrigerant holes **703** of the intermediate plate **70**, the plurality of refrigerant holes **503** of the upper cylinder **50**, and the plurality of upper refrigerant holes **913** of the upper flange **91**.

[0132] The refrigerant in the space between the upper flange **91** and the upper muffler **71** may move to the upper side of the upper muffler **71** through the plurality of openings **711** of the upper muffler **71**.

[0133] The flat muffler **72** may not include an opening or hole through which the refrigerant may be discharged. Therefore, the refrigerant discharged through the lower through hole **924** of the lower flange **92** may not be discharged into the oil in the oil reservoir **16** in which the flat muffler **72** is submerged through the flat muffler **72**.

[0134] The lower flange **92** may include a flange part **925**, a boss **926**, and a bearing **927**.

[0135] The flange part **925** may be formed in a disk shape. The flange part **925** may be formed to cover the hollow **61** of the lower cylinder **60**. The flange part **925** may be formed in a size corresponding to the lower cylinder **60**.

[0136] The flange part **925** may include the lower through-hole **924** communicating with the lower discharge port of the lower cylinder **60**. Therefore, the refrigerant discharged through the lower discharge port of the lower cylinder **60** may be discharged below the flange part **925** through the lower through-hole **924** of the flange part **925**.

[0137] The lower discharge valve **921** may be provided in the lower through-hole **924** of the flange part **925**. Therefore, the lower through-hole **924** of the flange part **925** may be opened and closed by the lower discharge valve **921**. When the refrigerant introduced into the lower cylinder **60** is compressed above a certain pressure, the lower discharge valve **921** may be opened so that the refrigerant may be discharged below the flange part **925**.

[0138] The boss **926** may extend vertically from the center of the flange part **925**. The boss **926** may extend vertically upward from the flange part **925**. A through hole may be formed in the center of the boss **926**.

[0139] The bearing **927** may be disposed in the through hole of the boss **926** and may rotatably support the drive shaft **30**. Accordingly, the drive shaft **30** may be rotatably supported by the bearing **927** and the upper bearing **917**. Any type of bearing may be used as the bearing **927** as long as it can rotatably support the drive shaft **30**. In the case of this disclosure, a sliding bearing may be used as the bearing **927**.

[0140] The plurality of first bolt holes **922** and the plurality of lower refrigerant holes **923** may be provided in the flange part **925** around the boss **926**.

[0141] The lower cylinder **60** may be provided with a plurality of bolt holes **602** corresponding to the plurality of first bolt holes **922** of the flange part **925**. In addition, the lower cylinder **60** may include the plurality of refrigerant holes **603** corresponding to the plurality of lower refrigerant holes **923** of the lower flange **92**.

[0142] The flat muffler **72** may be disposed on the lower surface of the flange part **925** of the lower flange **92**. Hereinafter, the flat muffler **72** will be described in detail with reference to FIGS. **8** and **9**.

[0143] FIG. **8** is a perspective view illustrating a flat muffler **72** used in a rotary compressor **1** according to an embodiment of the disclosure. FIG. **9** is a perspective view illustrating the flat muffler **72** of FIG. **8** as seen from above.

[0144] Referring to FIGS. **8** and **9**, the flat muffler **72** may be formed in an approximately dome shape and may include a plurality of concave portions **721** formed at regular intervals along the circumferential direction of the flat muffler **72** on the side surface of the flat muffler **72**. In detail, the flat muffler **72** may have a disk shape and the central portion of the disk may be formed to protrude downward in an approximately dome shape.

[0145] In addition, the flat muffler **72** may include a plurality of second bolt holes **722** formed at positions corresponding to the plurality of concave portions **721** along the edge of the flat muffler **72**. The plurality of second bolt holes **722** may be formed to correspond to the plurality of first bolt holes **922** of the lower flange **92**.

[0146] For example, the flat muffler **72** may include a fixing plate **723**, a caulking portion **724**, and a muffler portion **725**.

[0147] The fixing plate **723** may be formed in an approximately ring shape. The fixing plate **723** may include an outer circumferential surface formed in a circular shape and an inner circumferential surface concentric with the outer circumferential surface and having a diameter smaller than that of the outer circumferential surface. The inner circumferential surface of the fixing plate **723** may include a plurality of protrusions **7231**. The plurality of protrusions **7231** of the inner circumferential surface may be formed to protrude toward the center of the fixing plate **723**. The plurality of protrusions **7231** may be formed to correspond to the plurality of concave portions **721** of the muffler portion **725**. The plurality of protrusions **7231** may be formed in a curve corresponding to the plurality of concave portions **721** of the muffler portion **725**, respectively.

[0148] The portion between two adjacent protrusions **7231** may be concave to form a recess **7232**. In other words, the plurality of protrusions **7231** may form a plurality of recesses **7232**. Because each of the plurality of protrusions **7231** forms a projecting portion, the plurality of recesses **7232** and the plurality of projecting portions may be alternately provided on the inner circumferential surface of the fixing plate **723**. Accordingly, the inner circumferential surface of the fixing plate **723** may be formed concavely and convexly by the plurality of protrusions **7231**.

[0149] The fixing plate **723** may be fixed to the flange part **925** of the lower flange **92**. The fixing plate **723** may include the plurality of second bolt holes **722**. The plurality of second bolt holes **722** may be formed in the plurality of protrusions **7231** of the fixing plate **723**, respectively. The plurality of second bolt holes **722** may be formed to correspond to the plurality of first bolt holes **922** of the lower flange **92**.

[0150] The muffler portion **725** may be formed by extending downward from the inner circumferential surface of the fixing plate **723**. The muffler portion **725** may be formed to protrude in an approximately dome shape from the lower surface of the fixing plate **723**. The plurality of concave portions **721** may be formed at regular intervals in the circumferential direction on the side surface of the muffler portion **725**. The plurality of concave portions **721** may be formed as curved surfaces that protrude into the interior of the muffler portion **725**.

[0151] As illustrated in FIG. **9**, the plurality of concave portions **721** may protrude into the interior space of the muffler portion **725**. Therefore, the interior space of the muffler portion **725** may be

formed concavely and convexly.

[0152] A through hole **726** may be provided at the lower end of the muffler portion **725**. An oil pump disposed at the lower end of the drive shaft **30** may be immersed in the oil reservoir **16** through the through hole **726** of the muffler portion **725**.

[0153] The caulking portion **724** may be provided at the lower end of the muffler portion **725**. The caulking portion **724** may be provided at the edge of the through hole **726**. The caulking portion **724** may be formed to be fixed to one end of the boss **926** of the lower flange **92**. When the caulking portion **724** of the muffler portion **725** is joined to one end of the boss **926** of the lower flange **92**, the refrigerant may be prevented from leaking between one end of the boss **926** of the lower flange **92** and the through hole **726** of the muffler portion **725**.

[0154] The caulking portion **724** may be formed by performing a caulking operation along the entire circumference of the through hole **726** of the muffler portion **725**.

[0155] The muffler portion **725** may be provided between the fixing plate **723** and the caulking portion **724**.

[0156] The upper end of the flat muffler **72** may be formed not to protrude above the lower surface of the lower flange **92**. In detail, the fixing plate **723** of the flat muffler **72** may not protrude above the lower surface of the flange part **925** of the lower flange **92**, but may be positioned below the lower surface of the flange part **925**. In other words, the upper surface of the fixing plate **723** of the flat muffler **72** may be in contact with the lower surface of the flange part **925** of the lower flange **92**.

[0157] The upper surface of the fixing plate **723** of the flat muffler **72** may be formed not to be in contact with the side surface of the flange part **925** of the lower flange **92**. When the fixing plate **723** of the flat muffler **72** does not contact the side surface of the flange part **925** of the lower flange **92**, pressure may not be applied to the bearing **927** disposed in the lower flange **92** by the flat muffler **72**.

[0158] The sealing plate **100** may be disposed between the lower flange **92** and the flat muffler **72**. The sealing plate **100** may be formed to prevent or minimize the leakage of refrigerant between the lower flange **92** and the flat muffler **72**.

[0159] The sealing plate **100** may be positioned between the lower surface of the flange part **925** of the lower flange **92** and the upper surface of the fixing plate **723** of the flat muffler **72**.

[0160] FIG. **10** is a perspective view illustrating a sealing plate **100** used in a rotary compressor **1** according to an embodiment of the disclosure.

[0161] Referring to FIG. **10**, the sealing plate **100** may be formed in a flat plate having an approximately ring shape. The sealing plate **100** may be formed to correspond to the fixing plate **723** of the flat muffler **72**. The sealing plate **100** may be formed to seal the flat portion between the flat muffler **72** and the lower flange **92**.

[0162] The sealing plate **100** may include an outer circumferential surface formed in a circular shape and an inner circumferential surface concentric with the outer circumferential surface and having a diameter smaller than that of the outer circumferential surface.

[0163] The inner circumferential surface of the sealing plate **100** may include a plurality of bolt seats **101** and a plurality of refrigerant grooves **103**.

[0164] The plurality of bolt seats **101** may be formed to protrude from the inner circumferential surface of the sealing plate **100** toward the center of the sealing plate **100**. The plurality of bolt seats **101** may be formed to correspond to the plurality of protrusions **7231** of the fixing plate **723**. Accordingly, the plurality of bolt seats **101** may correspond to the plurality of concave portions **721** of the muffler portion **725**. The plurality of bolt seats **101** may be formed in a curve corresponding to the plurality of concave portions **721** of the flat muffler **72**, respectively.

[0165] A plurality of third bolt holes **102** corresponding to the plurality of second bolt holes **722** of the flat muffler **72** may be formed in the plurality of bolt seats **101**, respectively. In addition, the plurality of third bolt holes **102** of the sealing plate **100** may be formed to correspond to the

plurality of first bolt holes **922** of the lower flange **92**.

[0166] The plurality of refrigerant grooves **103** may be formed so that the sealing plate **100** does not cover the plurality of lower refrigerant holes **923** of the lower flange **92**. Therefore, when the sealing plate **100** is disposed on the lower surface of the lower flange **92**, the sealing plate **100** may not cover the plurality of lower refrigerant holes **923**.

[0167] The plurality of refrigerant grooves **103** may be formed between the plurality of bolt seats **101**. In other words, the plurality of refrigerant grooves **103** and the plurality of bolt seats **101** may be formed alternately in the circumferential direction of the sealing plate **100**.

[0168] Each of the plurality of bolt seats **101** may form a convex portion, and each of the plurality of refrigerant grooves **103** may form a concave portion. Accordingly, the inner circumferential surface of the sealing plate **100** may include a plurality of convex portions and a plurality of concave portions. In other words, the inner circumferential surface of the sealing plate **100** may be formed concavely and convexly. Accordingly, the inner circumferential surface of the sealing plate **100** may be formed concavely and convexly corresponding to the flat muffler **72**.

[0169] The sealing plate **100** may be formed of a material capable of preventing refrigerant from leaking between the flat muffler **72** and the lower flange **92**. For example, the sealing plate **100** may be formed of a heat-resistant resin, steel, copper, or the like. In addition, the sealing plate **100** may be formed of a material in which at least two of the heat-resistant resin, steel, and copper are laminated in a plate shape.

[0170] The flat muffler **72**, the sealing plate **100**, and the lower flange **92** may be fixed to the upper cylinder **50** by the plurality of bolts **93**. In detail, the plurality of bolts **93** may be fastened to the plurality of tap holes **502** of the upper cylinder **50** through the plurality of second bolt holes **722** of the flat muffler **72**, the plurality of third bolt holes **102** of the sealing plate **100**, the plurality of first bolt holes **922** of the lower flange **92**, the plurality of bolt holes **602** of the lower cylinder **60**, and the plurality of bolt holes **702** of the intermediate plate **70**. Then, the flat muffler **72**, the sealing plate **100**, the lower flange **92**, the lower cylinder **60**, and the intermediate plate **70** may be integrally fixed to the upper cylinder **50**.

[0171] The refrigerant discharged downward through the lower through hole **924** of the lower flange **92** may reduce in noise while passing through the inside of the flat muffler **72**, and then flow into the plurality of lower refrigerant holes **923** of the lower flange **92**.

[0172] The refrigerant introduced into the plurality of lower refrigerant holes **923** of the lower flange **92** may be discharged to the upper side of the upper flange **91** through the plurality of refrigerant holes **603** of the lower cylinder **60**, the plurality of refrigerant holes **703** of the intermediate plate **70**, the plurality of refrigerant holes **503** of the upper cylinder **50**, and the plurality of upper refrigerant holes **913** of the upper flange **91**.

[0173] The refrigerant discharged to the upper side of the upper flange **91** may be discharged to the upper side of the upper muffler **71**, that is, the space between the motor **20** and the compression part **40**, through the plurality of openings **711** of the upper muffler **71**.

[0174] The refrigerant that has moved to the space between the motor **20** and the compression part **40** may move to the upper side of the motor **20** through the motor **20**. For example, the refrigerant in the space between the motor **20** and the compression part **40** may move to the upper side of the motor **20** through the gap between the rotor **22** and the stator **21** and the plurality of refrigerant holes **27** provided in the rotor **22**.

[0175] The refrigerant that has moved to the upper side of the motor **20** may be discharged to the outside of the casing **10** through the refrigerant discharge pipe **14** disposed in the upper casing **11**.

[0176] Hereinafter, the positional relationship of the lower flange **92**, the flat muffler **72**, and the sealing plate **100** will be described in detail with reference to FIGS. **11** to **13**.

[0177] FIG. **11** is a view illustrating a state in which a sealing plate **100** and a flat muffler **72** are disposed in a lower flange **92**. FIG. **12** is a bottom view of FIG. **11**. FIG. **13** is a bottom view illustrating a state in which the flat muffler **72** is removed from FIG. **11**.

[0178] Referring to FIGS. **11** to **13**, the flat muffler **72** may be disposed on the lower surface of the lower flange **92**. The sealing plate **100** may be disposed between the lower flange **92** and the flat muffler **72**.

[0179] The sealing plate **100** may be disposed on the lower surface of the lower flange **92** so that the plurality of third bolt holes **102** thereof are aligned with the plurality of first bolt holes **922** of the lower flange **92**.

[0180] When the plurality of third bolt holes **102** of the sealing plate **100** are aligned with plurality of first bolt holes **922** of the lower flange **92**, the plurality of refrigerant grooves **103** of the sealing plate **100** may be positioned at the edges of the plurality of lower refrigerant holes **923** of the lower flange **92**.

[0181] At this time, the minimum distance between the inner circumferential surface of the sealing plate **100** and the plurality of lower refrigerant holes **923** of the lower flange **92** may be zero (0). For example, as illustrated in FIG. **13**, when the inner surface of the lower refrigerant hole **923** is in contact with the bottom surface of the refrigerant groove **103** of the sealing plate **100**, the distance between the lower refrigerant hole **923** and the inner circumferential surface of the sealing plate **100** may be zero (0).

[0182] Alternatively, the inner circumferential surface of the sealing plate **100** may be formed so that the bottom surfaces of the refrigerant grooves **103** of the sealing plate **100** are spaced apart from the inner surfaces of the lower refrigerant holes **923**. In this case, the distance between the lower refrigerant hole **923** and the inner circumferential surface of the sealing plate **100** may be greater than zero (0). At this time, the distance between the lower refrigerant hole **923** and the inner circumferential surface of the sealing plate **100** may be defined to be smaller than the minimum distance between the outer circumferential surface of the lower flange **92** and the lower refrigerant hole **923**. When the distance between the lower refrigerant hole **923** and the inner circumferential surface of the sealing plate **100** is greater than the minimum distance between the outer circumferential surface of the lower flange **92** and the lower refrigerant hole **923**, the refrigerant may leak.

[0183] In addition, in order to prevent refrigerant leakage, the distance between the bottom surface of the refrigerant groove **103** of the sealing plate **100** and the outer circumferential surface of the sealing plate **100** may be at least 1 mm. In other words, the distance between the concave portion of the inner circumferential surface of the sealing plate **100** and the outer circumferential surface of the sealing plate **100** may be at least 1 mm.

[0184] When the sealing plate **100** is formed with the structure as described above, the plurality of lower refrigerant holes **923** of the lower flange **92** may be exposed through the plurality of refrigerant grooves **103** of the sealing plate **100**. In other words, the plurality of lower refrigerant holes **923** of the lower flange **92** may not be covered by the sealing plate **100**.

[0185] Therefore, the refrigerant in the space between the lower flange **92** and the flat muffler **72** may flow into the lower refrigerant holes **923** of the lower flange **92** without interference of the sealing plate **100**.

[0186] In the case where the flat muffler **72** is disposed on the lower surface of the lower flange **92**, when the space between the lower flange **92** and the flat muffler **72** is filled with compressed refrigerant, as illustrated in FIG. **14**, a portion of the flat muffler **72** may be opened so that the refrigerant may leak therethrough.

[0187] FIG. **14** is a view illustrating a state in which a portion of a flat muffler **72** is opened by compressed refrigerant. For reference, in FIG. **14**, the opened portion is exaggerated to show the opened state of the flat muffler **72**.

[0188] Referring to FIG. **14**, reference numeral G indicates the opening of the flat muffler **72**. In other words, the opening of the flat muffler **72** refers to the vertical distance between the protrusion **7231** and the concave portion of the fixing plate **723**.

[0189] When the rotary compressor **1** operates so that refrigerant is discharged into the space

between the lower flange **92** and the flat muffler **72**, portions of the fixing plate **723** between the plurality of protrusions **7231** of the fixing plate **723** at which the plurality of bolts **93** that fix the flat muffler **72** to the lower flange **92** are disposed may be deformed by the pressure of the compressed refrigerant. In other words, the concave portion of the fixing plate **723** may be deformed downward from the protrusion **7231** thereof by the pressure of the refrigerant.

[0190] When there is no compressed refrigerant between the lower flange **92** and the flat muffler **72**, the plurality of concave portions of the fixing plate **723** of the flat muffler **72** may be restored to their original state and become flat with the plurality of protrusions **7231**. In other words, the opening G of the flat muffler **72** may be zero (0).

[0191] In order to minimize deformation of the bearing **927** of the lower flange **92** and to minimize the amount of refrigerant leaking between the lower flange **92** and the flat muffler **72**, the sealing plate **100** may be formed to have a certain thickness Ts. For example, the thickness Ts of the sealing plate **100** may be formed to have a certain thickness ratio with the thickness Tm of the flat muffler **72**.

[0192] The thickness Ts of the sealing plate **100** may be formed to have the following relationship with the thickness Tm of the flat muffler **72**: [0193] $0.5 \leq \text{thickness Ts of the sealing plate 100} / \text{thickness Tm of the flat muffler 72} \leq 2.0$

[0194] FIG. **15** is a graph illustrating a change in deformation of a bearing **927** and opening of a flat muffler **72** according to the thickness of a sealing plate **100**.

[0195] FIG. **15** illustrates the results of calculating the deformation of the bearing **927** disposed in the lower flange **92** and the opening G of the flat muffler **72** by computer simulation when the thickness Tm of the flat muffler **72** is 1.2 mm and the thickness Ts of the sealing plate **100** is 0.5 mm, 1 mm, 2 mm, 2.5 mm, and 3 mm. Here, the opening G of the flat muffler **72** refers to the vertical distance between the protrusion **7231** and the concave portion of the fixing plate **723** of the flat muffler **72**.

[0196] Referring to FIG. **15**, when the thickness Ts of the sealing plate **100** increases, the deformation of the concave portions between the plurality of protrusions **7231** of the fixing plate **723** of the flat muffler **72**, that is, the opening G of the flat muffler **72** may be reduced.

Accordingly, when the thickness Ts of the sealing plate **100** increases, the leakage of refrigerant between the flat muffler **72** and the lower flange **92** may be minimized or prevented.

[0197] In addition, when the thickness Ts of the sealing plate **100** increases, the deformation of the bearing **927** disposed in the lower flange **92** may increase. In detail, when the thickness Ts of the sealing plate **100** increases, the movement distance of the bolt **93** may increase, which may increase the stress generated in the boss **926** of the lower flange **92**. As the stress of the boss **926** increases, the deformation of the bearing **927** disposed in the boss **926** may increase.

[0198] When the deformation of the bearing **927** of the lower flange **92** increases, the input of the rotary compressor **1** may increase. When the input of the rotary compressor **1** increases, the energy efficiency (EER: energy efficiency ratio) may decrease.

[0199] When the deformation of the bearing **927** decreases, there is room for maintaining the oil film thickness, so that a design that reduces the diameter of the bearing is possible and the reliability of the rotary compressor **1** may be improved.

[0200] When a skirt type muffler is used, the bearing deformation is 10.5 μm . When the flat muffler **72** according to an embodiment of the disclosure and the sealing plate **100** having a thickness of 1 mm are used, the bearing deformation may be reduced to 6.7 μm .

[0201] Therefore, the thickness Ts of the sealing plate **100** may be defined so as to minimize the bearing deformation and the opening G of the flat muffler **72**.

[0202] Referring to FIG. **15** again, in the case where the thickness Tm of the flat muffler **72** is 1 mm, when the thickness Ts of the sealing plate **100** is 0.5 mm or less, the bearing deformation is 6.7 μm , which may not increase the input of the rotary compressor **1**, but the opening of the flat muffler **72** becomes 18.0 , which may cause excessive leakage of the

refrigerant.

[0203] When the thickness T_s of the sealing plate **100** is 2.5 mm or more, the opening G of the flat muffler **72** becomes 12.0 , so that the leakage of the refrigerant may be appropriate, but the bearing deformation is 7.3 , which may increase the input of the rotary compressor **1**. Therefore, the thickness T_s of the sealing plate **100** may be more than 0.5 mm and less than 2.5 mm.

[0204] For example, considering the ratio of the thickness T_s of the sealing plate **100** to the thickness T_m of the flat muffler **72**, when the thickness T_m of the flat muffler **72** is 1.2 mm, the thickness T_s of the sealing plate **100** may be 0.6 mm to 2.4 mm.

[0205] The performance of the rotary compressor **1** according to an embodiment of the disclosure having the above structure was compared with the performance of a rotary compressor according to the prior art using a skirt muffler.

[0206] Table 1 below is a performance comparison table of the rotary compressor **1** according to an embodiment of the disclosure and the rotary compressor according to the prior art using a skirt muffler. FIG. **16** is a graph illustrating a change rate in energy efficiency of a rotary compressor **1** according to an embodiment of the disclosure relative to the energy efficiency of a rotary compressor according to the prior art using a skirt muffler.

[0207] The comparative tests below were performed under ASHRAE-T conditions. ASHRAE-T conditions are condensing temperature of 54.4° C., liquid temp of 46.1° C., evaporating temperature of 7.2° C., suction temperature of 35° C., and ambient temperature of 35° C.

TABLE-US-00001

TABLE 1	operating cooling power (Btu/h)	input (W)	EER	speed related art
disclosure related art	disclosure related art	disclosure related art	disclosure related art	disclosure related art
30	23,906	+0.5%	2,058	-1.2%
11.61	+1.7%			
45	36,598	+0.4%	3,079	-0.6%
11.88	+1.1%			
60	49,947	+0.4%	4,246	-0.5%
11.76	+0.8%			

[0208] In Table 1, the prior art refers to a rotary compressor having a skirt muffler disposed on the lower flange **92**. The disclosure refers to the rotary compressor **1** having the flat muffler **72** and the sealing plate **100** disposed on the lower flange **92**. Here, the skirt muffler is formed to include a skirt that wraps around the side surface of the flange part **925** of the lower flange **92**. In other words, the skirt muffler includes the skirt that extends upward from the outer circumferential surface of the fixing plate **723** of the flat muffler **72** according to an embodiment of the disclosure. The unit of the operating speed is revolutions per second (rps). EER represents energy efficiency. Referring to Table 1, it can be seen that the rotary compressor **1** using the flat muffler **72** and the sealing plate **100** according to an embodiment of the disclosure has increased cooling power compared to the rotary compressor according to the prior art. In other words, the cooling power of the rotary compressor **1** using the flat muffler **72** and the sealing plate **100** according to an embodiment of the disclosure may increase by 0.4% to 0.5% compared to the cooling power of the rotary compressor according to the prior art. Accordingly, it can be seen that the refrigerant leakage between the flat muffler **72** and the lower flange **92** is minimized by the sealing plate **100**.

[0209] In addition, it can be seen that the rotary compressor **1** using the flat muffler **72** and the sealing plate **100** according to an embodiment of the disclosure has a reduced input compared to the rotary compressor according to the prior art. In other words, the input of the rotary compressor **1** using the flat muffler **72** and the sealing plate **100** according to an embodiment of the disclosure may be reduced by 0.5% to 1.2% compared to the input of the rotary compressor according to the prior art. Accordingly, it can be seen that the bearing deformation of the lower flange **92** due to the sealing plate **100** and the flat muffler **72** is minimized.

[0210] In addition, it can be seen that the rotary compressor **1** using the flat muffler **72** and the sealing plate **100** according to an embodiment of the disclosure has increased energy efficiency compared to the rotary compressor according to the prior art. In other words, the EER of the rotary compressor **1** using the flat muffler **72** and the sealing plate **100** according to an embodiment of the disclosure may increase by 0.8% to 1.7% compared to the EER of the rotary compressor according to the prior art. Accordingly, the rotary compressor **1** using the flat muffler **72** and the sealing plate

100 according to an embodiment of the disclosure may increase the EER by an average of 1.2% in the operating range of 30 to 60 rps compared to the rotary compressor according to the prior art. [0211] As described above, in the rotary compressor **1** according to an embodiment of the disclosure, the flat muffler **72** and the sealing plate **100** may be disposed on the lower flange **92**, so that the bearing deformation of the lower flange **92** may be minimized and the refrigerant leakage due to the flat muffler **72** may be minimized, thereby improving the energy efficiency. [0212] In the foregoing, the disclosure has been shown and described with reference to various embodiments. However, it is understood by those skilled in the art that various changes may be made in form and detail without departing from the scope of the disclosure as defined by the appended claims and equivalents thereof.

Claims

1. A rotary compressor comprising: a casing; a motor to be disposed inside the casing; a compression part to be disposed below the motor; a lower flange to be disposed below the compression part; a muffler, having a portion that is flat, to be disposed on a lower surface of the lower flange; and a sealing plate to be disposed between the muffler and the lower flange.
2. The rotary compressor of claim 1, wherein the portion that is flat is an upper end of the muffler does not protrude above the lower surface of the lower flange.
3. The rotary compressor of claim 1, wherein the muffler is formed in a dome shape and includes a plurality of concave portions formed at regular intervals along a circumferential direction of the muffler on a side surface of the muffler, and wherein the sealing plate is formed in a ring shape and includes an inner circumferential surface formed concavely and convexly corresponding to the muffler.
4. The rotary compressor of claim 3, wherein the lower flange includes a plurality of lower refrigerant holes and a plurality of first bolt holes formed along an edge of the lower flange, and wherein the muffler includes a plurality of second bolt holes formed corresponding to the plurality of first bolt holes of the lower flange at positions corresponding to the plurality of concave portions along an edge of the muffler.
5. The rotary compressor of claim 4, wherein the sealing plate includes a plurality of bolt seats formed to protrude toward a center of the sealing plate from an inner circumferential surface of the sealing plate and having a plurality of third bolt holes corresponding to the plurality of second bolt holes of the muffler.
6. The rotary compressor of claim 5, wherein the sealing plate is formed so as not to cover the plurality of lower refrigerant holes of the lower flange.
7. The rotary compressor of claim 6, wherein a distance between a concave portion of the sealing plate and an outer circumferential surface of the sealing plate is at least 1 mm.
8. The rotary compressor of claim 1, wherein a thickness of the sealing plate has a relationship with a thickness of the muffler as follows: $0.5 \leq \text{thickness } T_s \text{ of the sealing plate} / \text{thickness } T_m \text{ of the muffler} \leq 2.0$.
9. The rotary compressor of claim 1, wherein oil is received in a lower portion of the casing, and the muffler is immersed in the oil.
10. The rotary compressor of claim 1, wherein the sealing plate is formed of one of heat-resistant resin, steel, copper, and a material in which at least two of the heat-resistant resin, the steel, and the copper are laminated.
11. The rotary compressor of claim 1, wherein the lower flange comprises: a flange part formed in a disk shape; a boss extending vertically from the flange part and including a through hole; and a bearing disposed in the through hole of the boss, and wherein the muffler comprises: a fixing plate formed in a ring shape and fixed to the flange part of the lower flange; a caulking portion fixed to one end of the boss; and a muffler portion provided between the fixing plate and the caulking

portion.

12. The rotary compressor of claim 11, wherein the muffler portion includes a plurality of concave portions formed at regular intervals in a circumferential direction of the muffler.

13. The rotary compressor of claim 12, wherein the fixing plate includes a plurality of protrusions corresponding to the plurality of concave portions of the muffler portion.

14. The rotary compressor of claim 13, wherein the flange part of the lower flange includes a plurality of lower refrigerant holes and a plurality of first bolt holes, and the plurality of protrusions of the fixing plate includes a plurality of second bolt holes corresponding to the plurality of first bolt holes.

15. The rotary compressor of claim 14, wherein an inner circumferential surface of the sealing plate includes a plurality of bolt seats formed to correspond to the plurality of protrusions of the fixing plate and a plurality of refrigerant grooves formed so as not to cover the plurality of lower refrigerant holes of the flange part.
